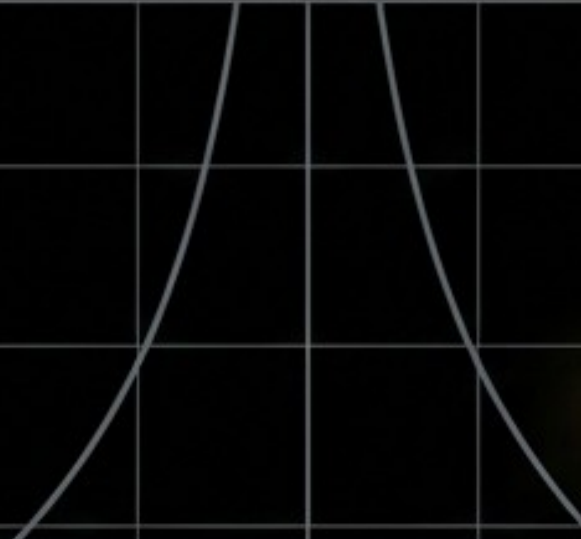


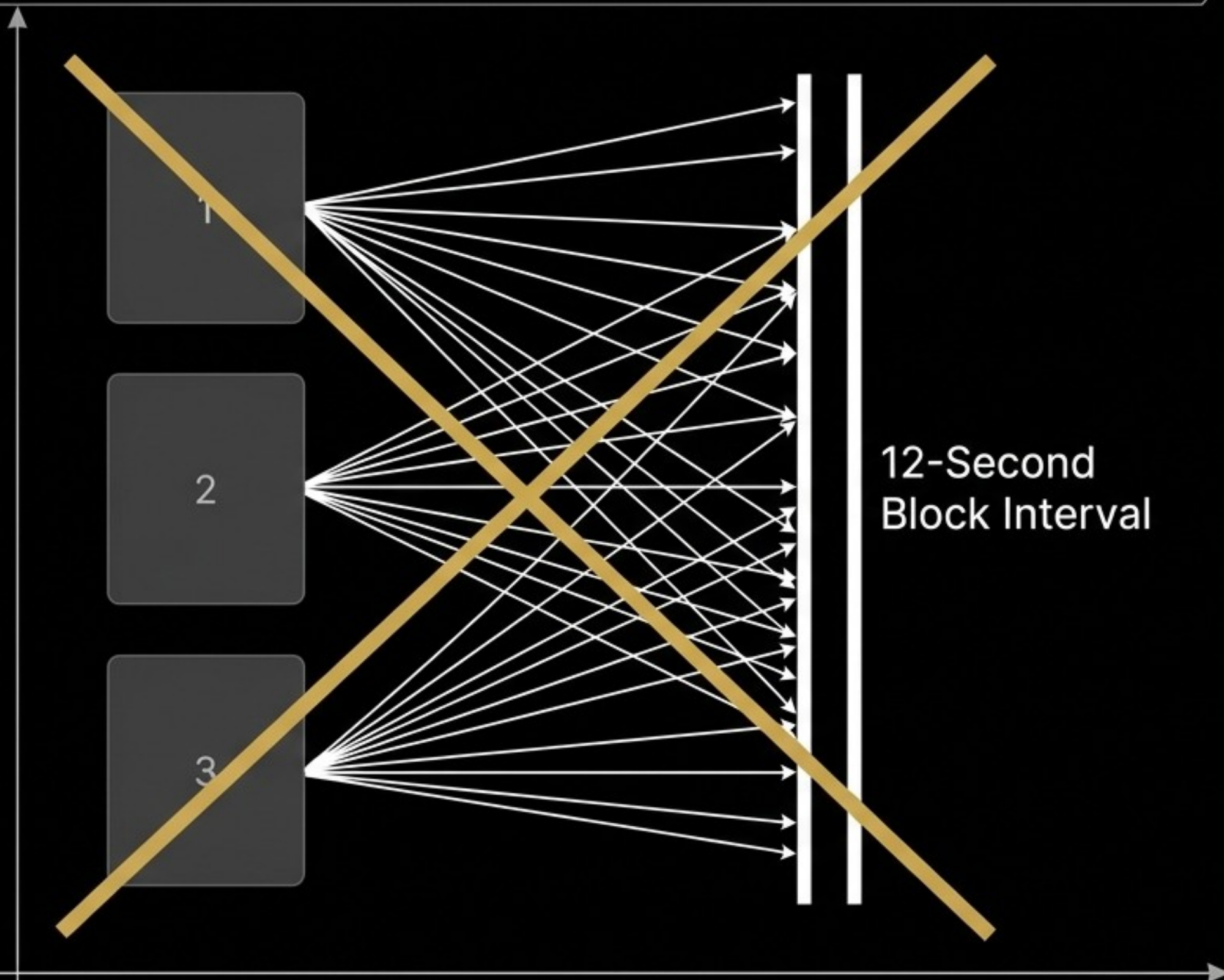

$$x \cdot y = k$$

AUTOMATED MARKET & LIQUIDITY POOLS

Fundamentals of Distributed Trust

CONSTANT PRODUCT INVARIANTS | SHARED LIQUIDITY MECHANICS
IMPERMANENT LOSS DYNAMICS | CONCENTRATED CAPITAL EFFICIENCY

The L1 Bottleneck: CLOB Failure



The Traditional Foundation

CLOB matches bids and asks in microseconds on centralized servers (e.g., NASDAQ, Binance), relying on institutional market makers to supply liquidity.

The On-Chain Reality

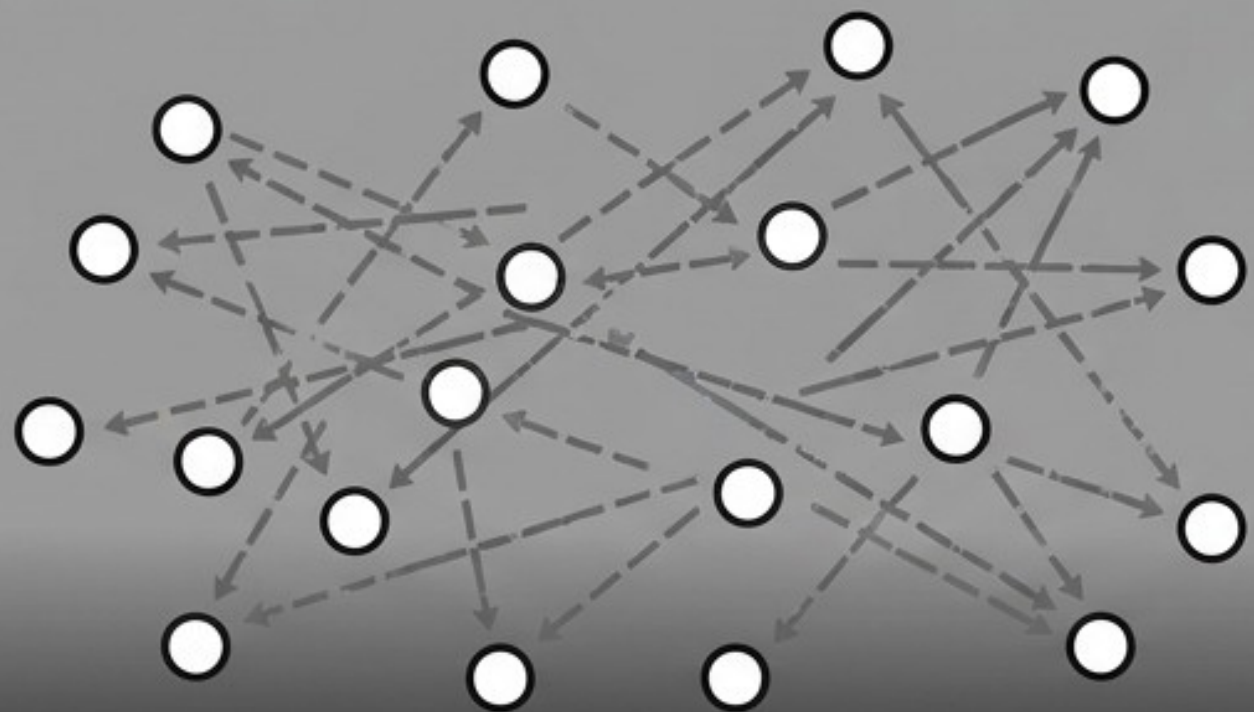
On a public blockchain, every order, cancellation, and price adjustment requires a distinct on-chain transaction.

The Fatal Flaw

12-second block intervals and fluctuating gas costs create a catastrophic bottleneck. Market makers are exposed to latency **front-running** and catastrophic gas bleed (e.g., the EtherDelta collapse of 2016).

1

Model CLOB (Peer-to-Peer)



- Requires precise bilateral coincidence of wants (identical price, volume, time).
- Passive liquidity is heavily fragmented across specific order book price levels.

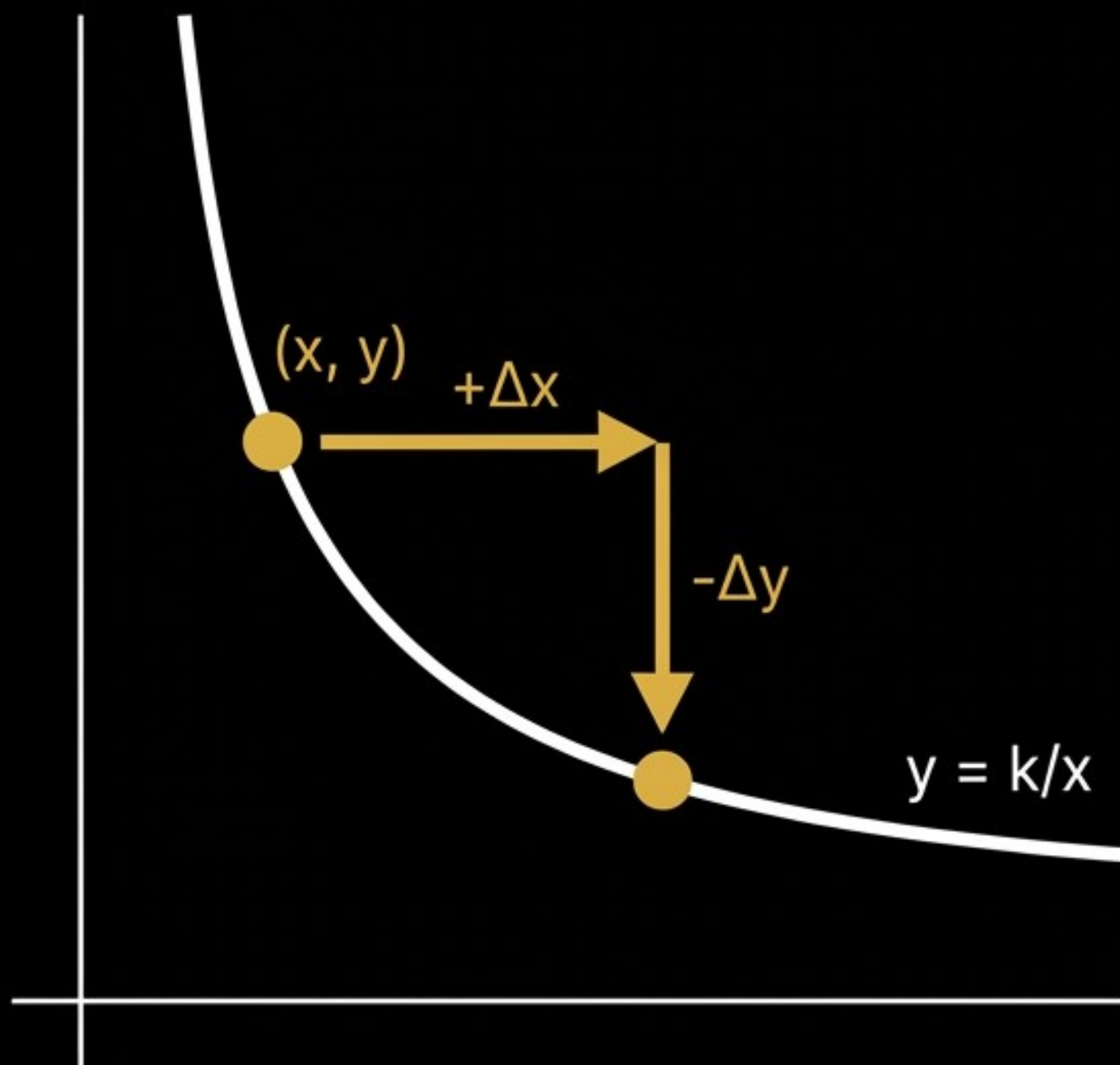
2

Model AMM (Peer-to-Contract)



- Eliminates direct matching. Traders interact with a deterministic, mathematically governed reserve of funds.
- Liquidity Providers (LPs) pool equivalent assets into a single smart contract, earning fees passively via automated invariant calculations.

The Mathematical Engine: $x \cdot y = k$



Constant Product Market Maker (CPMM)

The mathematical governance protocol of Uniswap V2.

Variables:

- **x**: Total reserve of Token X.
- **y**: Total reserve of Token Y.
- **k**: The invariant constant.

The Rule of Conservation:

The product of both reserves post-trade cannot be less than k . The pool mathematically never runs empty; marginal prices approach infinity as reserve balances approach zero.

Swap Mechanics & Fee Derivation

- ① Calculate effective input:

$$\Delta x_{effective} = 0.997 \cdot \Delta x$$

- ② Enforce constant product:

$$(x + 0.997 \cdot \Delta x)(y - \Delta y) = x \cdot y$$

- ③ Isolate output token:

$$\Delta y = \frac{0.997 \cdot \Delta x \cdot y}{x + 0.997 \cdot \Delta x}$$

Standard Fee Allocation

0.30% is strictly allocated to LPs, applying a multiplier of $\gamma = 0.997$ to the input.

Execution: This single, elegant formula is executed on-chain for every swap, guaranteeing deterministic output without an order book.

Market Dynamics: Price Impact vs. Slippage

Price Impact (Internal Curve Mechanics)

Definition: Deterministic price degradation caused by order size relative to total pool depth.

Mechanism: Moves directly along the $x \cdot y = k$ curve in a single transaction.

Example: Selling 10 ETH into a pool of 100 ETH & 300,000 USDC (Spot = \$3,000). The formula yields 27,198 USDC—an execution price of \$2,719. A brutal 9.3% deviation (\$2,801 loss) mathematically enforced by the curve.

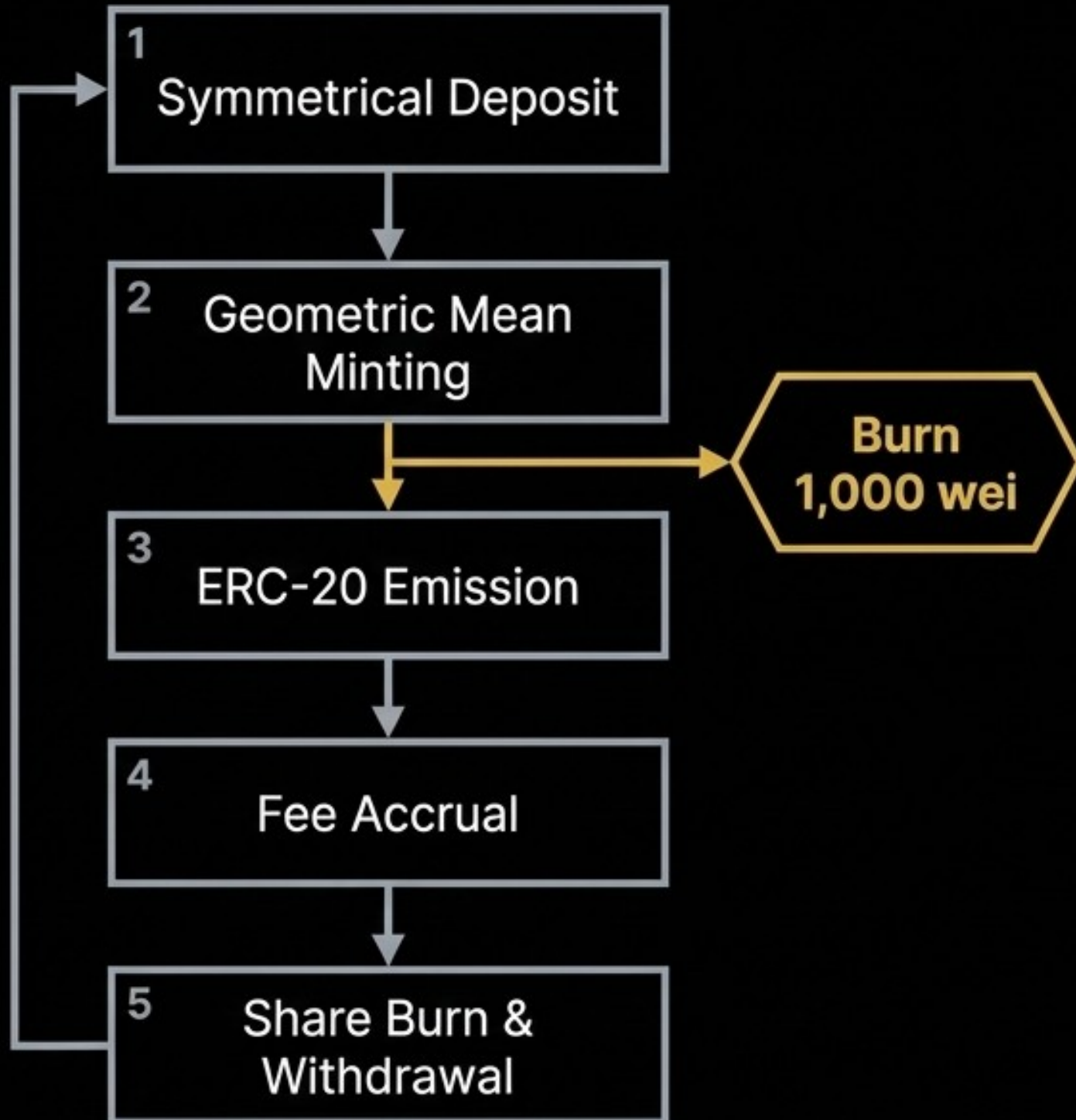
Slippage (External Latency Mechanics)

Definition: Latency-driven deviation between expected price and final block-validation price.

Mechanism: External MEV bots or rival traders execute in the mempool during the 12-second block window, shifting the starting point of the curve.

Mitigation: Wallet-level slippage tolerance limits (triggers automatic transaction revert).

LP Architecture: Minting & Burn Protection



LP Share Tokens

Smart contracts mint unique ERC-20 tokens representing proportional ownership of total pool reserves.

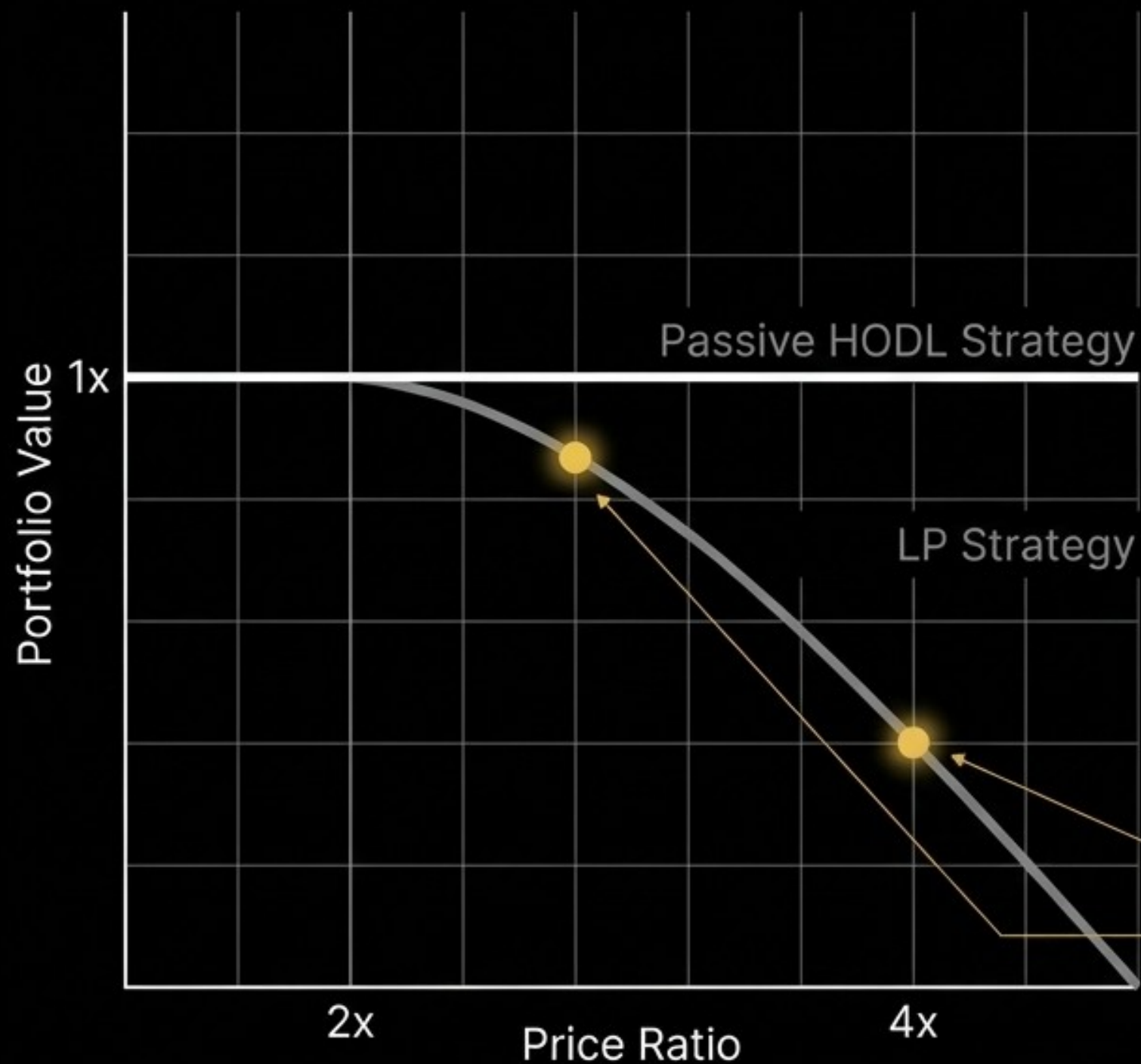
Initialization

Initial supply is minted via geometric mean:
$$S_{\text{minted}} = \sqrt{(x_0 \cdot y_0)} - 1,000.$$

First Depositor Defense

Uniswap V2 permanently burns the first 1,000 wei of shares to the zero address (0x000...000). This permanently prevents First Depositor Inflation Attacks, neutralizing truncation exploits on empty pools.

The Risk Protocol: Impermanent Loss (IL)



Definition:

The portfolio value deficit of providing liquidity versus passively holding assets in a cold wallet.

Arbitrage Extraction:

As external markets surge, pool prices lag. Arbitrageurs extract appreciating assets at a discount until internal prices match global parity, leaving LPs holding depreciating assets.

The Formula:

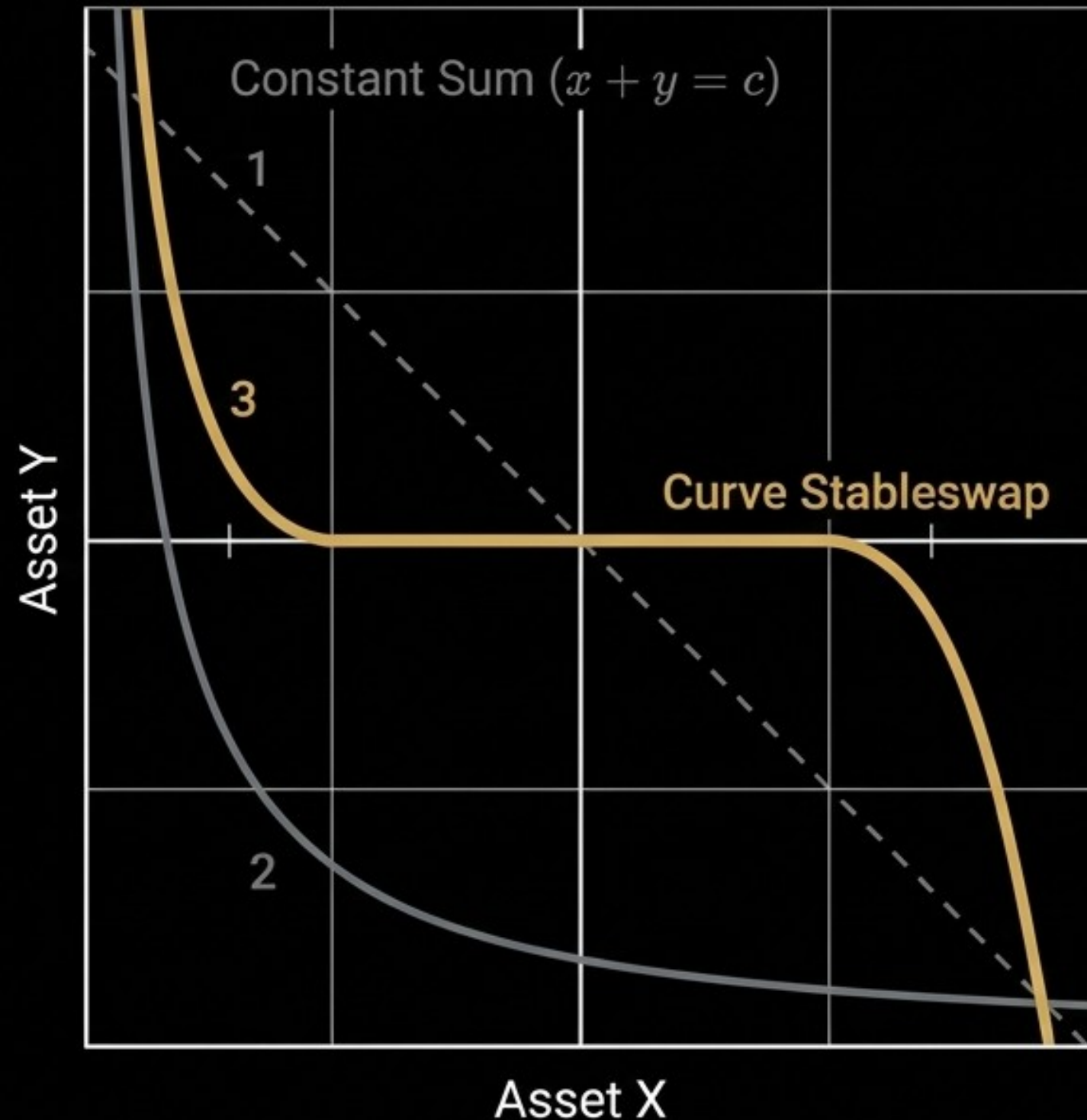
$$IL(r) = (2\sqrt{r}) / (1 + r) - 1$$

Relative Scale:

- 2x price surge = -5.7% IL
- 4x price surge = -20.0% IL

Note: Loss becomes 'permanent' only upon withdrawal while prices remain divergent.

Curve Stableswap: Bridging The Invariants



1. The Stablecoin Dilemma

Constant Product creates catastrophic price impact for 1:1 pegged assets (USDC/USDT). Constant Sum offers zero slippage but drains entirely during de-peg events.

2. The Hybrid Solution

Curve Finance (2019) fuses both geometric curves using a dynamic amplification coefficient (A).

3. Adaptive Dynamics

Delivers massive, zero-slippage liquidity around the 1:1 parity zone, but dynamically bends into a protective hyperbola during extreme de-peg crises to secure pool reserves.

Uniswap V3: Concentrated Liquidity

V2



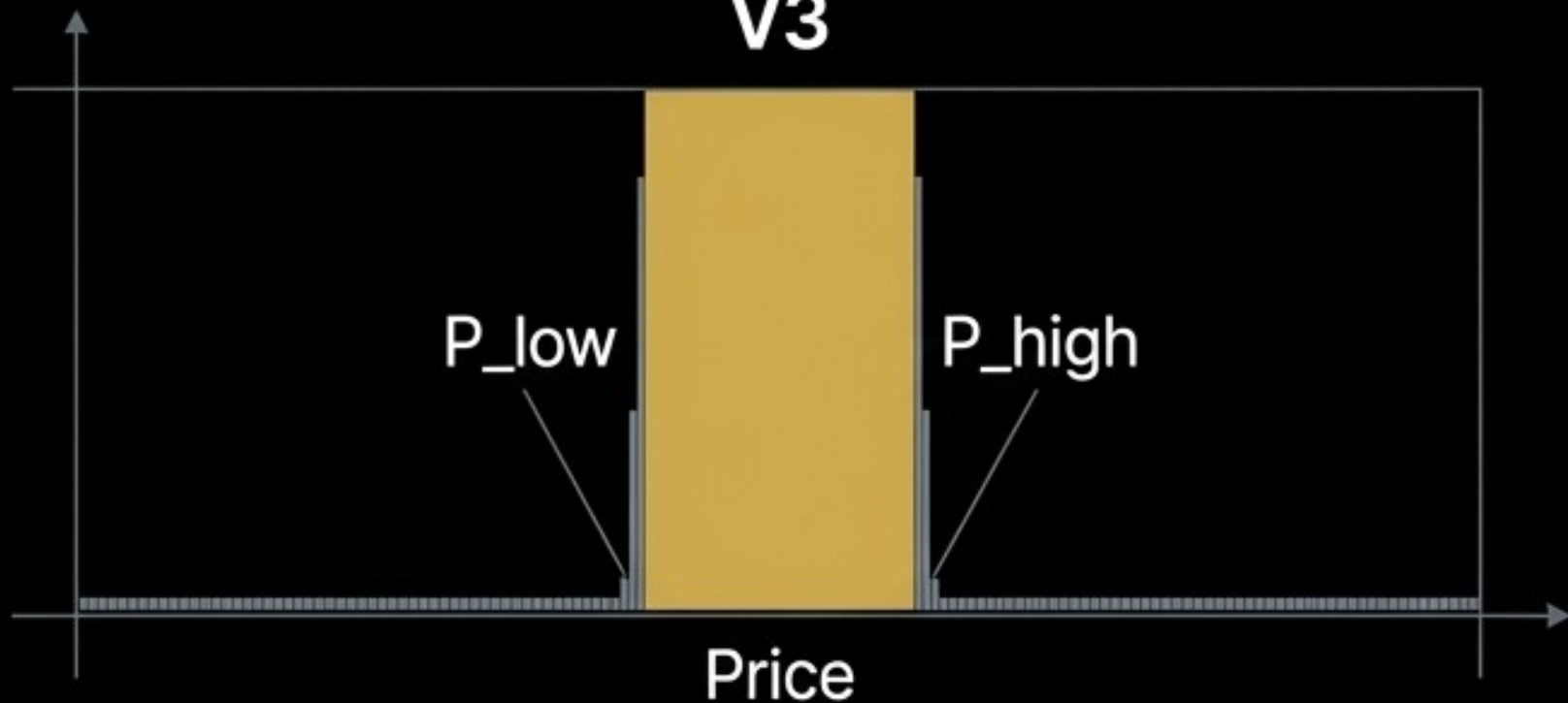
V2 Capital Inefficiency

Uniform liquidity spread means >90% of LP capital idles in unreachable price extremes (\$1 or \$1,000,000) and generates zero fees.

V3 Concentrated Liquidity

LPs allocate capital exclusively within custom price boundary “ticks” ($[P_{low}, P_{high}]$).

V3



The Result

Up to 4,000x capital efficiency on narrow ranges.

Architectural Shift

Because LP parameters are now custom and non-fungible, liquidity positions are minted as unique ERC-721 NFTs, not fungible ERC-20s. Out-of-range positions cease generating fees.